ANALYZING RAINFALL TRENDS AND LANDSCAPE FEATURES: A STUDY OF AMBEDKAR NAGAR DISTRICT

Project Report

Submitted to the

Acharya Narendra Deva University of Agriculture and Technology

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महामाया कृषि अभियांत्रिकी एवं प्रौद्योगिकी महाविद्यालय,
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(A.N.D. University of Agriculture and Technology, Kumarganj, Ayodhya)

Date. 18/05/2024

CERTIFICATE-I

This is to certify that the project report entitled **"ANALYZING RAINFALL TRENDS AND LANDSCAPES FEATURES: A STUDY OF AMBEDKAR NAGAR DISTRICT"** submitted by Mr. HIMESH TIWARI (E-11592/20), Mr. KISHAN SINGH (E-11593/20), Miss. VIPRA MISHRA (E-11604/20) under my supervision and guidance and the results embodied in this report have not been submitted to any other university for the award of any degree or diploma. The assistance and help received during the course of investigation have been fully acknowledged.

Ambedkar Nagar
May, 2024

Dr. Vikas Kumar Singh
Advisor

महामाया कृषि अभियांत्रिकी एवं प्रौद्योगिकी महाविद्यालय,
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CERTIFICATE-II

This is to certify that the project report entitled **"ANALYZING RAINFALL TRENDS AND LANDSCAPES FEATURES: A STUDY OF AMBEDKAR NAGAR DISTRICT"** submitted by Mr. HIMESH TIWARI (E-11592/20), Mr. KISHAN SINGH (E-11593/20), Miss. VIPRA MISHRA (E-11604/20) to the Acharya Narendra Deva University of Agriculture and Technology, Kumarganj, Ayodhya, (U.P.) for the partial fulfillment of award of degree of **"Bachelor of Technology (Agricultural Engineering)"** has been approved by the advisory committee and examiner after an oral examination.

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TABLE OF CONTENT

Sr. No./Chapters	Particulars	Page No.
1.	CERTIFICATE-I	II
2.	CERTIFICATE-II	III
3.	ACKNOWLEDGEMENT	IV
4.	TABLE OF CONTENTS	VI
5.	LIST OF FIGURES	VIII
6.	LIST OF TABLES	IX
7.	LIST OF NOMENCLATURE/ SYMBOL	X
Chapter 1	**INTRODUCTION**	1
1.1	Rainfall	1
1.2	Trend analysis	2
1.3	Sen's Slope estimator test	3
1.4	Morphometric analysis of drainage basin or watershed	3
1.5	Types of morphometric analysis	3
Chapter 2	**REVIEW OF LITERATURE**	9
2.1	Trend Analysis	9
2.2	Morphometric Analysis	12
Chapter 3	**MATERIALS AND METHODS**	19
3.1	Study area	19
3.2	Climate	20
3.3	Soil type	20
3.4	Data Specification	20
3.5	Methodologies	20

Chapter 4	**RESULTS AND DISCUSSION**	26
4.1	Trend Analysis	26
4.2	Morphometric Analysis	32
Chapter 5	**SUMMARY AND CONCLUSIONS**	36
	REFERENCES	37
	ABSTRACT	41
	BIO-DATA	42

LIST OF FIGURES

Fig. No.	Particulars	Page No.
3.1	Description of Study Area	19
4.1	Monthly Z statistics	27
4.2	Seasonal and Annual Z statistic	27
4.3	Decadal analysis of Z statistics	28
4.4	Seasonal and annual analysis in decade of Z statistics	29
4.5	Streams of Maharuwa micro-watershed	32

LIST OF TABLES

Table No.	Particulars	Page No.
1.1	Morphometric characteristics parameter	7
3.1	Numerical value of watershed	25
4.1	Heat map of 40 years Sen's value	29
4.2	Heat map of 40 years seasonal and annual Sen's value	30
4.3	Heat map of decadal analysis	31
4.4	Heat map of seasonal and annual decadal analysis	31
4.5	Stream order and number of stream order	33
4.6	Morphometric parameters and their values	35

LIST OF NOMENCLATURE/ SYMBOL

Symbol	Description
%	Percentage
&	Ampersand
^{o}	Degree
cm	Centimeter
<	Less than
et al.	et alibi and others
Fig.	Figure
i.e.	That is
km	Kilometer
ha	Hectare
mm	Millimeter
dia	Diameter
m	Meter
m^2	Square meter
m^3	Cubic meter
no.	Number
GIS	Geological Information System
>	Greater than
avg.	Average
Eq^n	Equation

Chapter 1
INTRODUCTION

This chapter introduce rainfall trend analysis and watershed morphometric analysis to understand climate variability and watershed characteristics.

1.1 Rainfall

Rainfall is a meteorological phenomenon characterized by the precipitation of liquid water droplets from the atmosphere onto the Earth's surface. It plays a crucial role in sustaining ecosystems, agriculture, and human societies. The process of rainfall formation is influenced by various atmospheric conditions, including temperature, humidity and atmospheric pressure, as well as factors such as wind patterns, topography and the presence of cloud-forming nuclei. The rainfall patterns of Ambedkar Nagar district, Uttar Pradesh, hold significant importance due to their implications for agriculture, water resource management, and local livelihoods. Understanding the temporal and spatial variability of rainfall in this region is crucial for mitigating risks associated with droughts, floods, and crop failures.

This chapter aims to comprehensively analyse the rainfall characteristics of Ambedkar Nagar District, focusing on factors influencing rainfall variability, trends over time, and the impact on agricultural productivity. By examining historical data and employing statistical and spatial analysis techniques, this study seeks to provide valuable insights into the dynamics of rainfall in the region and contribute to informed decision-making for sustainable development. The rainfall patterns and monsoon seasons vary depending on the region. In many tropical and subtropical regions, there are typically three main monsoon seasons:

1.1.1 Pre-monsoon season

The pre-monsoon season, spanning March, April, and May, brings rising temperatures and increased humidity. This period often features thunderstorms and dust storms, setting the stage for the upcoming monsoon rains that rejuvenate the parched landscape.

1.1.2 Monsoon season

This is the main rainy season, typically occurring in the summer months. It brings the 75% of the annual rainfall. The monsoon season, occurring in June, July, August, and September delivers heavy rainfall and cooler temperatures. This vital period replenishes water sources, supports agriculture, and transforms the landscape

into lush greenery, while sometimes causing floods and landslides due to the intense downpours.

1.1.3 Post-monsoon season

The late monsoon season, spanning October and November, features lingering rains and a gradual drop in temperatures. This period is crucial for sustaining water levels and supporting late agricultural cycles. The retreating monsoon's gentler rains signify the transition from the intense monsoon to the drier winter months.

1.1.4 Winter Season

The northeast monsoon season, occurring in December, January, and February, brings rain primarily to India's southeastern coast. This period replenishes water reserves, supports winter crops, and moderates temperatures. The rains, though less intense than the southwest monsoon, are vital for agriculture and maintaining ecological balance

1.2 Trend Analysis

The purpose of trend analysis is to determine the patterns of rainfall are generally rising or falling over the period of time. According to Duhan and Pandey (2013), rainfall trend analysis is the process of using statistical techniques to determine the pattern of changes in a long-term rainfall dataset over a given time frame. (Mann, 1945: Basistha et al., 2009: Oguntunde et al., 2011) states that the Mann-Kendall test is a non-parametric test for finding patterns in data series and statistic distribution for testing on non-linear trends and turning points. In numerous studies, the Mann-Kendall test has been effectively applied to ascertain statistically significant or insignificant trends in hydrologic climatic variables, such as precipitation, within the framework of climate change. Experiments without any presumptions about the population being studied are called non-parametric tests.

Mann-Kendall test as well as modified Mann-Kendall test are generally used to analyse the trend of a particular set of data using time series data for each variable over a significant period, ensuring regular intervals and addressing any missing values or outliers. Ranking the data points in chronological order, Mann-Kendall statistic is utilized for measurement of the strength and direction of trends. Positive statistics indicate increasing trends, while negative values suggest decreasing trends.

Significance testing compares the computed statistic to critical values, determining if the trend is statistically significant. Interpretation of results involves assessing whether trends exist and their directionality, considering implications for

climate patterns and environmental conditions. Additionally, visualizations such as time series plots aid in understanding the trends over time, providing valuable insights into the variability of rainfall.

1.3 Sen's Slope Estimator Test

In the context of trend analysis of rainfall, Sen's slope estimator plays a vital role in assessing the long-term changes in precipitation patterns over the time. Sen's slope estimator provides a robust method for estimating the trend in rainfall data, particularly in areas where rainfall variability and extreme events are common. To apply Sen's slope estimator in trend analysis of rainfall, typical historical rainfall data is needed to collect over a period of time, such as monthly or annual measurements. Then, using Sen's method, median slope of all possible pairs of data points is calculated, representing the change in rainfall intensity over time. This median slope represents the overall trend in rainfall, whether it's increasing, decreasing, or remaining stable.

1.4 Morphometric Analysis of Watershed

Morphometric analysis of watersheds is to understanding their hydrological behaviour, geomorphic characteristics, and environmental significance. This chapter focuses on comprehensive morphometric study of the watershed aiming to delineate its spatial organization, quantify morphometric parameters, and assess their implications for water resource management and ecological sustainability. By employing geospatial techniques and statistical methods, this study endeavours to provide a holistic understanding of the watershed morphological attributes and contribute to informed land use planning and conservation efforts.

1.5 Types of Morphometric Analysis

Morphometric analysis mainly divided in three aspects and their names are-

1. Linear aspect (One-Dimensional)
2. Areal aspect (Two-Dimensional)
3. Relief aspect (Three-Dimensional)

1.5.1 Linear aspects

The linear aspect of a watershed refers to its physical characteristics in terms of its length, shape, and the flow of water within its boundaries.

1.5.1.1 Stream order

Stream order is a hierarchical classification system that categorizes streams and rivers based on their position within a drainage network. The concept of stream order

was introduced by Robert E. Horton in 1945. The smallest, first-order streams have no tributaries, while second-order streams are formed by the confluence of two first-order streams, third-order streams by the confluence of two second-order streams, and so on. The stream order increases as tributaries join the main streams, creating higher-order streams. Stream order is valuable for studying river network organization, watershed delineation, and understanding hydrological processes.

1.5.1.2 Stream length

Stream length refers to the distance measured along the main channel of a river or stream from its source to its mouth, where it typically empties into a larger body of water such as a lake, ocean, or another river. Stream length is a fundamental parameter for assessing the size and spatial extent of river systems. It provides valuable information for watershed management, floodplain mapping, and understanding river basin dynamics. Stream length can vary widely depending on factors such as topography, geology, climate, and human activities.

1.5.1.3 Mean stream length

Mean stream length is a statistical measure calculated by dividing the total length of all stream segments within a river basin or watershed by the total number of stream segments. It represents the average length of individual stream reaches within the drainage network. Mean stream length is useful for characterizing the spatial distribution of stream lengths and understanding the connectivity and branching patterns of river networks. It can provide insights into landscape evolution, sediment transport, and ecological processes in riverine ecosystems.

1.5.1.4 Bifurcation ratio

The ratio of links or stream segments of one order to those of the next higher order.

1.5.1.5 Stream length ratio

Stream length ratio is the ratio of mean length of streams of one order to that of the next lower order stream segments, which tends to be constant throughout the successive orders of a watershed and has an important relationship with the discharge of surface flow and erosional stages of the basin.

1.5.2 Aerial aspects

The areal aspect of a watershed pertains to its spatial characteristics, focusing on its area, shape, boundaries, and topographic features.

1.5.2.1 Drainage density

Drainage density is a measure of the total length of streams or channels within a drainage basin relative to its total area. It is calculated by dividing the total length of all streams (including both main channels and tributaries) by the area of the drainage basin. Drainage density provides insights into the degree of stream branching and channel development within a watershed, with higher values indicating a denser network of channels and potentially higher runoff potential.

1.5.2.2 Stream frequency

Stream frequency represents the density of streams or channels within a defined area. It is calculated by dividing the total number of stream segments (including both main channels and tributaries) by the area of interest. Stream frequency is often expressed as the number of streams per unit area (e.g., streams per square kilo-meter) and provides information about the spatial distribution and connectivity of stream channels within a watershed.

1.5.2.3 Texture ratio

Texture ratio quantifies the degree of channel branching or stream order distribution within a river network. It is calculated by dividing the total length of higher-order streams (e.g., second-order, third-order, etc.) by the total length of all streams within the drainage basin. Texture ratio values close to unity indicate a balanced distribution of stream orders, while values significantly different from one suggest either a dominance of lower-order streams (e.g., dendritic drainage pattern) or higher-order streams (e.g., trellis or parallel drainage patterns).

1.5.2.4 Form factor

Form factor is a dimensionless parameter that describes the compactness or shape complexity of a drainage basin. It is calculated by dividing the area of the drainage basin by the square of its perimeter. Form factor values closer to zero indicate more elongated or irregular basin shapes, while values closer to one suggest more compact or circular shapes. Form factor provides insights into the degree of basin elongation and geometric simplicity.

1.5.2.5 Circulatory ratio

Circulatory ratio quantifies the degree of circularity or roundness of a drainage basin. It is calculated by dividing the area of the basin by the square of its perimeter multiplied by 4π (the theoretical area of a perfect circle with the same perimeter). Circulatory ratio values close to one indicate nearly circular or rounded basin shapes,

while values less than one indicate more elongated or irregular shapes. Circulatory ratio helps characterize the geometric regularity of drainage basins.

1.5.2.6 Elongation ratio

Elongation ratio quantifies the degree of elongation or length-to-width ratio of a drainage basin. It is calculated by dividing the maximum length of the basin (from its headwaters to its outlet) by its maximum width (measured perpendicular to the maximum length). Elongation ratio values close to one indicate more symmetrical or equidimensional basin shapes, while values greater than one indicate elongated or elongated basin shapes. Elongation ratio provides insights into the aspect ratio and geometric elongation of drainage basins.

1.5.2.7 Length of overland flow

Surface runoff and overland flow are two quite distinct hydrological phenomena. Overland flow is the flow of precipitated water that travels across the surface of the ground and into stream channels. Surface runoff is the channel flow that reaches the watershed's exit. While runoff is more important in larger watersheds, overland flow is more important in smaller ones. The overland flow is shallow both in depth and length. Laminar conditions characterise the overland flow characteristics.

1.5.3 Relief aspects

In a watershed, relief aspects refer to the variation in elevation and topographic features such as hills, valleys, and slopes, influencing water flow and hydrological processes.

1.5.3.1 Basin relief

Basin relief refers to the difference in elevation between the highest and lowest points within a drainage basin or watershed. It provides a measure of the vertical relief or elevation range across the landscape enclosed by the drainage divide. Basin relief is calculated by subtracting the elevation of the basin outlet (typically where the river exits the basin) from the elevation of its highest point (often a peak or ridge). Basin relief is an important parameter for assessing the geomorphic complexity, terrain ruggedness, and hydrological behaviour of watersheds.

1.5.3.2 Relief ratio

Relief ratio, also known as basin relief ratio or hypsometric integral, is a dimensionless parameter that quantifies the degree of relief or elevation variability within a drainage basin relative to its total area. It is calculated by dividing the basin relief (the elevation difference between the highest and lowest points) by the total area

of the basin. Relief ratio values close to zero indicate relatively flat or low-relief landscapes, while higher values indicate more rugged or mountainous terrain. Relief ratio provides insights into the morphological characteristics, erosion processes, and geological history of watersheds.

1.5.3.3 Ruggedness number

Ruggedness number, also known as the Terrain Ruggedness Index (TRI) or Roughness Index, is a quantitative measure of terrain ruggedness or surface roughness within a drainage basin. It is calculated based on the variability in elevation across the landscape, typically using elevation data from Digital Elevation Models (DEMs). Ruggedness number is computed as the mean absolute difference in elevation between each cell and its surrounding cells within a defined neighbourhood or moving window. Higher ruggedness numbers indicate more rugged or irregular terrain, while lower values correspond to smoother or flatter landscapes. Ruggedness number is widely used in geomorphological analysis, landform classification, and environmental modelling to characterize terrain complexity and identify areas of high relief or topographic variability.

Table1.1: Morphometric characteristics parameter

Aspects	Morphometric Parameters	Methods
Linear aspect	Stream order (U)	Hierarchical order
	Stream length (L_u)	Length of the stream
	Mean stream length (L_{sm})	$L_{sm} = L_u/N_u$ where, L_u=Stream length of order 'U' N_u=Total number of stream segments of order 'U'
	Stream length ratio (R_l)	$R_l = L_u/L_{u-1}$ where L_u=Total stream length of order 'U', L_{u-1}=Stream length of next lower order.
	Bifurcation ratio (R_b)	$R_b = N_u/N_{u+1}$ where, N_u=Total number of stream segment of order 'u'; N_{u+1}=Number of segments of next higher order
Areal aspect	Basin relief (B_h)	Vertical distance between the lowest and highest points of watershed.
	Relief ratio (R_h)	$R_h = B_h/L_b$ Where, B_h=Basin relief; L_b=Basin length
	Ruggedness number (R_n)	$R_n = B_h \times D_d$ Where, B_h=Basin relief; D_d=Drainage density

Relief aspect	Drainage density (D_d)	$D_d = L/A$ where, L=Total length of streams; A=Area of watershed
	Stream Frequency (Fs)	$Fs = N/A$ where, N=Total number of streams; A=Area of watershed
	Texture ratio (T)	$T = N1/P$ where, N1=Total number of first order streams; P=Perimeter of watershed
	Form factor (R_f)	$R_f=A/(L_b)^2$ where, A=Area of watershed, Lb=Basin length
	Length of overland flow (Lof)	$Lof=1/2D_d$ where, D_d is the drainage density

Source: Rekha et al., (2011)

1.5.4 Justification

Over half of India's population still makes their living from agriculture and the Indian agriculture primarily depends on rainfall. In the state of Uttar Pradesh, agriculture is the primary source of income for the livelihood. Water is a vital natural resource that is necessary for life to exist. The population is growing, which is driving up water demand. Water is necessary for industry, agriculture, human, and animal consumption. For this reason, it is crucial to manage this vital resource in a sustainable manner. Where runoff is extremely high due to different topographical circumstances, proper management and development planning are required to restore or recharge the water. Keeping all these aspects in mind this study is planned with the following objectives:

Objectives

1. To analyse the trend and magnitude of trend of rainfall for Ambedkar Nagar, Uttar Pradesh.
2. To estimate the Morphometric parameters of Maharuwa micro-watershed located in Ambedkar Nagar, Uttar Pradesh.

Chapter 2
REVIEW OF LITERATURE

Numerous studies have been previously carried out on the trend analysis of rainfall and estimation of the morphometric parameters of a watershed. The available works were reviewed and presented here on the following aspects.

1. Trend analysis.
2. Estimation of the morphometric parameters.

2.1 Trend analysis

Trend analysis of rainfall by Mann-Kendall and Sen's Slope test delves into seminal studies applying these methodologies in climate science and water resource management. It scrutinizes their effectiveness, limitations, and advancements, providing insights into the evolving landscape of rainfall trend analysis. By synthesizing past research, it aims to contextualize and contribute to the understanding of long-term precipitation patterns and their implications

Chung et al., (2000) interpreted recent trends observed in temperature and precipitation over Korea at Korea-China Centre for Atmospheric Research, Korea. Meteorological data and satellite observations suggest that changes have occurred in the characteristics of the quasi-stationary fronts that produce summer rain. In recent years scattered local heavy showers usually occur with an inactive showery front, in comparison with the classical steady rain for more than three weeks. For instance, local heavy rainfall, on 6 August 1998 was in the range of 123 mm. The scattered convective storms resulted in flooding with a heavy toll of approx. 500 people.

Arora et al., (2005) evaluated the trends of temperature over India, at Indian Institute of Technology, Roorkee, India. The impact of climate change is projected to have different effects within and between countries. Information about such change is required at global, regional and basin scales for a variety of purposes. An investigation was carried out to identify trends in temperature time series of 125 stations distributed over the whole of India. In the research three variables related to temperature, viz. mean, mean maximum and mean minimum. Temperature anomalies are plotted, and it is observed that annual mean temperature, mean maximum temperature and mean

minimum temperature have increased at the rate of 0.42, 0.92 and 0.09°C per 100 years, respectively.

Jain et al., (2012) analysed the rainfall and temperature trends for northeast India at Department of Civil Engineering, National Institute of Technology, Silchar, Assam, India. Trend analysis of rainfall data series for 1871–2008 did not show any clear trend for the region as a whole, although there are seasonal trends for some seasons and for some hydro-meteorological subdivisions. The maximum and average temperatures in the region were showing a rising trend and the increase was notably significant in statistical terms for the post-monsoon season (November and December). This behaviour of the data indicated the presence of an element of seasonal cycle in the temperature trends over NE India.

Jain and Kumar, (2013) analysed the rainfall and temperature trend for India at Department of Water Resources Development and Management, Indian Institute of Technology, Roorkee, India. Basin-wise trend analysis; 15 basins was decreasing trend in annual rainfall; only one basin showed significant decreasing trend at 95% confidence level. In a recent study, 6 river basins have shown increasing trend in annual rainfall in the 0.27–10.16 mm/yr range whereas 15 river basins have shown decreasing annual rainfall in the 0.45–4.93 mm/yr range. Further, 4 river basins have shown increasing trend in rainy days and 15 river basins have shown the opposite trend.

Swain et al., (2015) studied the statistical monthly rainfall trend analysis for Raipur district at Dept. of Civil Engineering, National Institute of Technology, (NIT) Raipur India. Trend analysis of monthly rainfall data for period of 102 years i.e. 1901 to 2002 was used. Reveal a significant decrease for the months of Southwest monsoon i.e. June, July, August and September, thereby inferring for a consequent decrease in annual rainfall. Reveals downward trend for most of the months of a year for the period under investigation. Since the months of Southwest monsoon i.e. June-September shows a significant decreasing trend, it can be inferred for the annual rainfall over Raipur to be decreasing.

Yunsheng et al., (2017) analysed and predicted the rainfall trends over Bangladesh using Mann–Kendall, Spearman's Rho tests and Aurto Regressive Integrated Moving Average (ARIMA) model at Department of Geography and Environmental Studies, University of Chittagong, Bangladesh. Sixty years monthly

rainfall data of Bangladesh were analysed to detect trends. Modified Mann–Kendall, Spearman's Rho tests and Sen's slope estimators were applied to find the long-term annual, dry season and monthly trends. The modified Mann– Kendall trend test results showed a very good agreement with the Spearman's rho test results which indicates that both the test methods are almost equally powerful to test rainfall time series trends.

Palaniswami et al., (2018) investigated the rainfall and temperature change point detection and trend analysis of series over Vellar river basin at Centre for Water Resources, College of Engineering, Anna University, Guindy, India. The magnitude of increasing trends in North East Monsoon (NEM) and South West Monsoon (SWM) are greater than that of summer and winter seasons for almost all the rain gauge stations. The maximum temperature and minimum temperature change points are identified in the years 1985 and 2001 and 1987 and 2013, respectively. Different non-parametric tests were used for trend analysis and change point detection. The results show that all monthly, seasonal, and annual maximum, minimum, and mean temperatures have a significant greater rising trend.

Eblaghian et al., (2019) analysed the trend of temperature, precipitation, and relative humidity Changes in Iran at Hydrology and Water Resource Engineering Department of Shahid Chamran University of Ahvaz, Iran. In the pursuit of detecting the trend and the shift in trend in hydro-meteorological variables, various statistical methods have been developed and used over the years. Two methods commonly used (parametric and non-parametric), the non-parametric method has been favoured over parametric methods. Long term trend analysis can reveal the beginning of the trend year, trend changes over time, and abrupt trend detection in a time-series. It is expected that the findings of this study will bring about more insights on understanding the regional hydrologic behaviour over the last several decades in Iran.

Ilori and Ajayi, (2020) investigated the future temperature and rainfall change detection and trend analysis of over West Africa. It examined future trends with change detection in temperature and rainfall over three agro-climatic zones of West Africa. Standard normal homogeneity, Buishand's, Pettitt's, and Mann–Kendall test was used for change point detection and trend analysis at 5% significant level. For the near-future, an increase in the mean temperature between 0.5–1.30 ℃ and 0.19–1.67 ℃ is projected to occur under RCP 4.5 and RCP8.5 respectively. Projected relative change in seasonal cycle shows that winter months may witness increase in rainfall amounts

under RCP4.5 but significantly dry under RCP8.5 in near and far-future as temperature is expected to become warmer in all months.

Pandit et al., (2021) studied the rainfall, temperature and relative humidity trend analysis of rainfall at department of physics, St. Xavier's College, Maitighar, Kathmandu, Nepal. Five years (2012-2016) data of Rainfall, Relative Humidity and Temperature from Tribhuvan International Airport station were analyzed. Nepal lies in the subtropical region in which experience four seasons in a year i.e., winter, spring, summer and autumn. The relation between Rainfall vs Relative Humidity and Relative Humidity vs temperature was found to be insignificant respectively throughout all five years. The relation between Rainfall and Temperature was found to be significant throughout all five years.

Dubey et al., (2023) conducted the statistical trend analysis of major climatic factors over Chhattisgarh state, India. A regression analysis has also been done to validate the obtained results from Sen's slope. The negative slope obtained from the regression analysis indicated a downward trend in annual rainfall similar to Sen's slope. The results showed a negative (downward) trend in rainfall for most of the months except for January, March, April, and September where an increasing (positive) trend is obtained while the trend in the month of July and the monsoon season is significantly decreasing as the obtained p-value is less than α (5% significance level i.e., 0.05). The temperature data analysis showed an increasing trend for most of the months except for January, May, and June while the months of September, October, and November and annual temperature are significantly increasing in the state of Chhattisgarh.

2.2 Morphometric analysis

Analysing the morphological features of the drainage basin helps to understand its hydrological characteristics and the association of water, with soil, topography, and vegetation of the catchment. Morphometric analysis reveals the linear, areal, and relief aspects of a drainage basin.

Siddiqui (2004) carried out land use studies focusing on mapping the past and present conditions and the extent of forests and rangelands using remote sensing and GIS technologies. These technologies provided a possible means of monitoring and mapping the changes occurring in natural resources and the environment on a continuous basis and have been used to evaluate the geographic extent and distribution

of the forest and to monitor temporal changes in the forest cover between specified years.

Chopra et al., (2005) computed morphometric parameters using remote sensing and GIS techniques. Sub-watersheds were analysed within the study area, and it covers over 94.49 km² and 12.58 km², while the perimeter of the area was 19.09 km and 5.74 km. Both the sub-watersheds had the highest 5th order stream. The sub-watersheds have a mean stream length of 1.28 km and 9.45 km. Values for form factor (0.26 and 0.38), circulatory ratio (0.44 and 0.67), and elongation ratio (0.58 and 0.70) were observed.

Pankaj and Kumar, (2009) researched on five major sub-watersheds of Song basin at Dehradun district in Uttarakhand for analysing morphometric parameters using GIS software. Watershed area is located in tropical, sub-tropical, and humid climate between 30°00′ to 30°30′ North latitude and 77°45′ to 78°15′ East longitude. Bifurcation ratio was found as 2.644, 2.009, 2.213, 1.839, and 1.813, and form factors of 0.121, 0.117, 0.134, 0.319, and 0.215 respectively of five different sub-watersheds. The circulatory ratio was determined as 0.300, 0.528, 0.603, 0.437, and 0.351. The elongation ratio was found to be 0.393, 0.385, 0.637, 0.523, and 0.413. The drainage patterns were found nearly dendritic to sub-dendritic. All five sub-watersheds had 5th to 6th order.

Sreedevi et al., (2009) analysed the Morphometric analysis of a watershed of South India Using SRTM data and GIS. An attempt has been made to study drainage morphometry and its influence on hydrology of Wailapalli watershed, South India. For detailed study we used Shuttle Radar Topographic Mission (SRTM) data for preparing Digital Elevation Model (DEM), aspect grid and slope maps, Geographical Information System (GIS) was used in evaluation of linear, areal and relief aspects of morphometric parameters. The study reveals that the elongated shape of the basin is mainly due to the guiding effect of thrusting and faulting. The mean Rb of the entire basin is 3.89 which indicate that the drainage pattern is not much influenced by geological structures.

Pareta et al., (2011) conducted the quantitative morphometric analysis of a watershed of Yamuna basin, India using Advanced Spaceborne Thermal Emission and Reflection Radio Meter (ASTER) and GIS. ASTER data was used for preparing Digital Elevation Model (DEM), and Geographical Information System (GIS) was used in

evaluation of linear, areal and relief aspects of morphometric parameters. Watershed boundary, flow accumulation, flow direction, flow length and stream ordering were prepared using Arc Hydro tool; and contour, slope-aspect, hill shade were prepared using Surface Tool in ArcGIS-10 software, and ASTER (DEM).

Mishra et al., (2011) evaluated the morphometric analysis of Tons basin, Rewa District, Madhya Pradesh, based on watershed approach. Ilwis 3.4 software was used drainage map of the study area. The stream numbers, orders, lengths and other morphometric parameters like bifurcation ratio, drainage density, stream frequency, shape parameters etc. were measured. The low values of bifurcation ratio (mean value of 3.6) and drainage density (mean value of 0.71) suggest that the area was not much affected by structural disturbances. The drainage basin study suggests that most of the rocks in the area are of quite permeable nature and there are good chances of infiltration.

Patel et al., (2012) performed the study on water harvesting structures location by using Geo-visualization concepts based on morphometric analysis. Study area was Varekhadi watershed in Surat district of Gujarat sprawled over 1500 km² area bounded by latitudes 21° 00″ to 21° 30″N and longitudes 73° 15″ to 73° 45″E. The watershed was divided into 26 sub watersheds of which all linear, aerial and relief aspects were analysed. The basin length 23.18 km and 4.14 km were determined by GIS tool for Varekhadi watershed. The texture ratio varied from 0.24 to 5.67. The elongation ratio was from 0.6 to 1.0 for the regions of low relief whereas value between 0.6 to 0.8 for high relief and areas.

Panhalkar et al., (2012) conducted the morphometric analysis and watershed development prioritization of Hiranya Keshi basin in Maharashtra, India. Land, water and soil are limited natural resources and their wide utilization with increasing population is a major area of concern. To mitigate the demand and supply gap between resources and ever-increasing demand, it is of prime importance to conserve the natural resources with proper prioritization for its sustainable development. Morphometric analysis of a watershed provides a quantitative description of a drainage system. It is an important aspect of the watershed characterization.

Magesh et al., (2013) determined morphometrical aspects of Bharatha puzha river basin located in Kerala and Tamil Nadu states. The drainage basin was spatially sprawled over an area of 5988.56 km2. SRTM-DEM (30-meters) data was processed

with geo processing tool. The perimeter and basin length were found as 569.04 km and 153.8 km respectively. Total relief and relief ratio were found as 2.504 km and 0.0163 respectively. Circulatory ratio and form factor were found as 0.23 and 0.25 respectively. The basin slope varies from 0 degree to 70.59 degrees with elongation ratio of 0.57. The drainage patterns in the basin were found dendritic to sub-dendritic.

Sreenivasulu (2013) conducted study as an analysis on land use land cover using remote sensing and GIS in and around Vempalli in Kadapa district of Andhra Pradesh, India. Geographic Information System (GIS) provides an effective tool for analysis of land use land cover changes at a regional level. The geospatial technology of RS and GIS holds the potential for timely and cost-effective assessment of natural resources. Under this study three maps such as location map, drainage map and land use land cover maps were developed. The land use and land cover analysis on the study area has been attempted based on thematic mapping of the area consisting of built-up land, cultivated land, water bodies, forest and uncultivated land using the satellite image. The research concludes that there is a rapid expansion of built-up area. With the help of Geographic Information System, the various land use and land cover zones are mapped, which in turn helps for decision maker for planning purpose.

Aruchamy and Kavitha, (2014) studied thematic mapping engine using GIS Technology. This project aimed to provide data, visualization tools and a cartographic tool box to the user in a web-based interface. The core subject of the study in which the thematic mapping engine was developed to demonstrate how these techniques could be utilized in a web application. A key component to this approach is the utilization of GIS technology.

Waikar et al., (2014) studied the morphometric Analysis of a drainage basin using Geographical Information System a case study at Civil and water management SGGS Institute of Engineering and Technology, Vishnupuri, Nanded. The morphometric parameters of basin can address linear, areal and relief aspects. The present study deals mainly with the geometry, more emphasis being placed on the evaluation of morphometric parameters such as stream order (Nu), stream length (Lu), bifurcation ratio (Rb), drainage density (D), stream frequency (Fs), texture ratio (T), elongation ratio (Re), circularity ratio (Rc) and form factor ratio (Rf) etc. Study area is Charthana, geographically located between 76° 30′, 76° 40′ E longitudes, and 19°30′, 19°45′ N latitudes located in Parbhani district of Maharashtra state in India.

Pande et al., (2017) inspected the GIS based quantitative morphometric analysis and its consequences: a case study from Shanur river basin, at department of Geo informatics, Shir Shivaji College, Maharashtra, India. The extracted drainage network was classified according to Strahler's system of classification and it reveals that the terrain exhibits dendritic to sub-dendritic drainage pattern. Hence, from the study, it is concluded that remote sensing data (SRTM–DEM data of 30 m resolution) coupled with geo processing techniques prove to be a competent tool used in morphometric analysis and evaluation of linear, slope, areal and relief aspects of morphometric parameters. The combined outcomes have established the topographical and even recent developmental situations in basin.

Verma and Jha, (2017) studied extraction of watershed characteristics using GIS and Digital Elevation Model. This study shows generation of digital elevation model (DEM) and extraction of geo-morphological characteristics from DEM using GIS. Aim of this study was to extract watershed characteristics with the help of GIS using DEM. Hence the work was carried out for the preparation of DEM using contour lines of the topographic maps related to the study area and delineation of watershed and sub-watershed boundaries along with delineation of drainage lines.

Chandniha and Kansal., (2017) worked on the prioritization of sub-watersheds relies on morphometric analysis using GIS technique of Piperiya watershed, India. Study watershed is originated from Hasdeo catchment in Mahanadi basin in Chhattisgarh state of India. Watershed has total area of 2,414 km2 with highest and lowest elevation of 1,062 m and 324 m respectively. Total number of streams and stream length of all order were found 1,264 and 1,589 km, respectively. Study revealed that the majority of first order i.e., about 87% with 5.92 and 3.10 bifurcation ratios for SWS-1 and SWS-7 as the highest and smallest, respectively.

Meshram and Sharma., (2017) worked on Prioritization of watershed through morphometric parameters by Principal Component Analysis (PCA-based approach). Study watershed runs along the Shakhar river rises in the Satpura range, east of the chindi village at Chhindwara district in the Madhya Pradesh with elevation of 600 m. The watershed covers about 2220 km2.The Shakhar river catchment was classified into 8 sub-watersheds had basin area of 538.22 km2 and 9.23 km2, basin perimeter of 171.78 km and 17.03 km as highest and smallest respectively. The bifurcation ratio was found highest 5.52 for sub-watershed-6 while lowest 3.49 for sub-watershed-1.

Relative reliefs were found as highest for sub-watershed-3 valued 0.030 whereas 0.007 for sub-watershed-2.

Sukristiyanti et al., (2018) studied the watershed-based morphometric analysis a review at Research Center for Geo technology, Indonesian Institute of Sciences, Indonesia. Morphometric analysis of watershed is the best method to identify the relationship of various aspects in the area. Despite many technical papers were dealt with in this area of study, there is no particular standard classification and implication of each parameter. This paper deals with the meaning of values of the various morphometric parameters, with adequate contextual information. A critical review is presented on each classification, the range of values, and their implications. Besides classification and its impact, the authors also concern about the quality of input data, either in data preparation or scale/the detail level of mapping.

Mangan et al., (2019) inspected the morphometric analysis of watershed using remote sensing and GIS—a case study of Nanganji river basin in Tamil Nadu, India at Geographic Information Systems, NIIT University, Neemrana, Rajasthan, India. Nanganji river basin is one of the least studied seasonal river basins in India which carry substantial flows during the monsoon period. Morphometry of Nanganji river basin, located in the central Tamil Nadu prairies, has been studied using remote sensing and GIS. The interrelationship between the various morphometric factors of the basin has been studied using a correlation matrix. Finally, the study identifies environmental issues of the Nanganji river basin mostly related to the river flow regime which widens significantly during the monsoon months.

Khalkho et al., (2019) studied preparation of various thematic maps for Dhangaon micro watershed of Bemetara district. The satellite data of IRS –P6 LISS-IV of 10th October 2014 of the study area were used for generating various thematic maps such as land use land cover, soil, hydro geomorphology and slope map. Remote Sensing and GIS technique can be effectively used for the development of various thematic maps such as land use land cover, soil series, hydro geomorphology, slope, drainage etc. The land use land cover map indicates that major part of the study area is dominated by paddy crop (199.37 ha) which is 33% of the total area. This is followed by Soybean (124.07 ha), which is 21.54%current fallow (113.08 ha) which is 19.64% of the total area pigeon pea (54.439 ha) which is 9.45% of the total area. Deep water body contributes 2.15% of the total area and small water body by 0.015% of the total

area. Four classes of soil texture were identified with inceptisols (sandy clay loam) contribute 309.84 ha, which is60.86%of the total area followed by alfisol covering 161.826 ha, along with vertisols and entisols covering (72.69 ha) and (46.74 ha) respectively.

Bogale (2021) studied the morphometric analysis of a drainage basin using geographical information system in Gilgel Abay watershed, Lake Tana basin, upper Blue Nile Basin, Ethiopia. The present study addressed linear and areal morphometric aspect of the watershed. The study deals with emphasis on the evolution of morphometric parameters such as stream order, stream length, bifurcation ratio, drainage density, stream frequency, texture ratio, elongation ratio, circularity ratio, and form factor ratio. The morphometric analysis of the basin revealed that Gilgel Abay is firth-order drainage basin with total of 662 drainage network, of which 511 are first order, 111 are second order, 30 are third order, 9 are fourth order, and 1 is fifth-order stream.

Salunke et al., (2023) studied the morphometric Analysis of Panjhara river basin with use of GIS for development of watershed plan Department of Civil and Environmental Engineering, Mumbai, India. The research demonstrates that, for improved planning and maintenance, GIS and RS data may be utilized to analyze and approximately measure the duration and pace of erosional activities in a Panjhara river basin. The technique uses a 30-meter Shuttle Radar Topography Mission Digital Elevation Model (SRTM-DEM) and survey of India toposheet for efficient and quick extraction of morphometric data and hypsometric analysis.

Marwade et al., (2024) studied the mapping and prioritizing flash-flood susceptible watersheds in the Warana basin, India. A morphometric analysis and total ranking approach for resilience planning at department of geography, savitribai pule Pune university, Pune. Delineated 13 sub-watersheds and employed GIS techniques to directly measure basic parameters from the DEM, while others were calculated using mathematical formulas. The sub watersheds of the Warana basin were prioritized using the total rank method. The morphometric analysis revealed that the Warana basin was a sixth-order basin with a total area of nearly 2085 km^2, a length of 104.75 km, and a perimeter of 460 km.

Chapter 3

MATERIALS AND METHODS

This chapter outlines the materials and methods employed in conducting trend analysis of rainfall as well as morphometric analysis of Maharuwa micro-watershed located in Ambedkar Nagar, Uttar Pradesh. It describes the statistical techniques employed, such as Mann-Kendall and Sen's slope test for rainfall trend analysis, and morphometric analysis tools for watershed characterization, providing a systematic framework for the study.

3.1 Study Area

The present study area is located in the Ambekar Nagar district of Uttar Pradesh. The study area lies between 26°09'N and 26°40' N latitude and 82°12'E and 83°05' E longitude. It spans an area of approximately 2,520 square kilometers and is situated in the fertile Gangetic plains (Anon., 2024a). The district is characterized by diverse topography, ranging from flat plains to undulating terrain, with an average elevation of 88 meters above sea level that is shown in figure 3.1(Anon., 2024b). It is primarily agrarian, with agriculture being the primary occupation of the population.

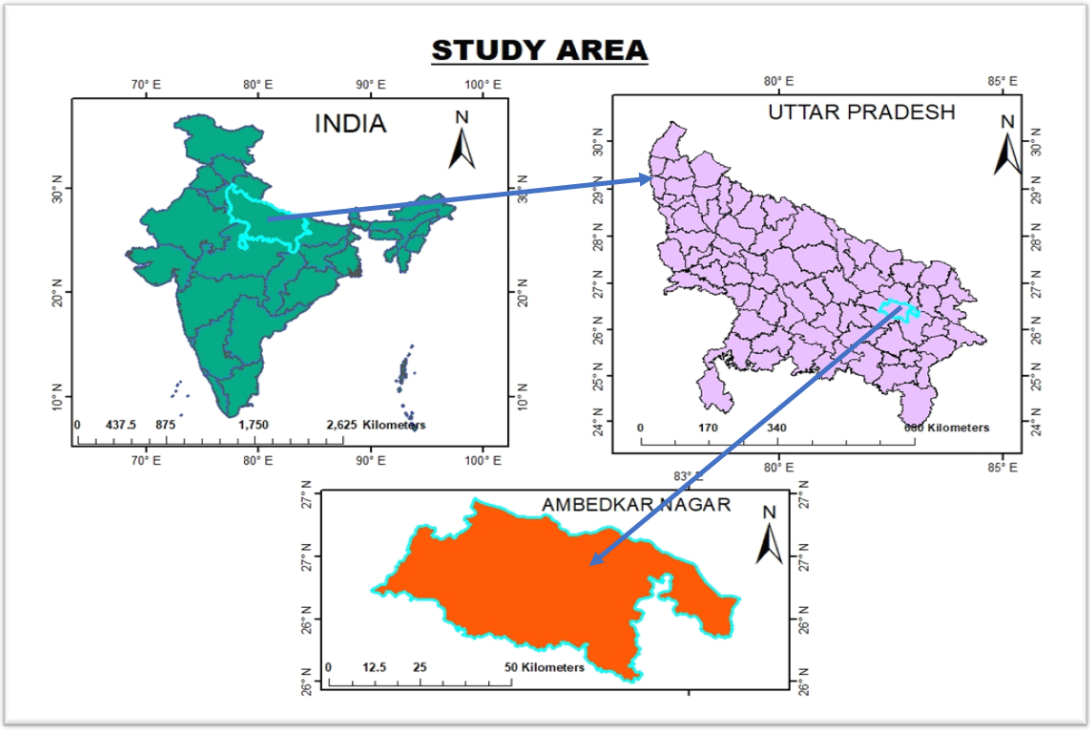


Fig. 3.1 Description of study area

3.2 Climate

Ambedkar Nagar experiences a subtropical climate, with hot summers, monsoonal rainfall, and cool winters, making it an ideal location for studying rainfall trends and watershed morphology. Ambedkar Nagar is characterized by its predominantly rural landscape, with agriculture being the primary occupation of the local population. The district is endowed with fertile soil and experiences a subtropical climate, with hot summers and mild winters. Major landforms include plains and gentle slopes, with the Rapti river flowing through the region.

3.3 Soil types

Soils of Ambedkar Nagar encompass various types, predominantly alluvial soils along with sandy, clayey, and loamy soils. Each soil type has distinct characteristics that affect its water-holding capacity, infiltration rates, and susceptibility to erosion. The soil composition of Ambedkar Nagar plays a crucial role in regulating surface runoff and groundwater recharge, thereby influencing local rainfall patterns. Alluvial soils, with their high permeability, facilitate rapid infiltration of rainfall into the ground, reducing surface runoff and enhancing groundwater recharge. Conversely, areas with clayey soils may experience surface waterlogging during intense rainfall events due to their low permeability.

3.4 Data Specification

This study utilizes a comprehensive dataset consisting of 40 years of rainfall data sourced from a Water Resource Information System (WRIS) spanning from 1980 to 2019. The morphometric analysis of the Maharuwa micro-watershed is conducted using SRTM (DEM) file data. This analysis quantitatively evaluates with the help of GIS which measure the watershed physical characteristics such as its drainage network, shape, and relief aspects.

3.5 Methodologies

An extensive explanation of the methods employed for the proposed study is given in this section.

3.5.1 Trend Analysis

To assess the temporal changes, the trend analysis method was applied. It is divided into two different types: confirmatory and exploratory data analysis. It is made

up of several graphical, nonparametric, and parametric statistical approaches. The Mann Kendall Test was used for trend analysis. We can determine if the observed values of a variable have grown or shrunk over time by using trend analysis.

3.5.1.1 Mann-Kendall test

The non-parametric Mann-Kendall (MK) test has been used to determine whether there is a statistically significant trend in hydrologic climatic variables, such as temperature and precipitation, with regard to climate change. The MK test compares the alternative hypothesis that there is an increasing or declining trend with the null hypothesis, which states there is no trend. The Modified Mann-Kendall (MMK) test is a statistical method used to detect trends in time series data. It is an adaptation of the Mann-Kendall test, which is a non-parametric test used to determine if there is a monotonic trend in a dataset over time. The Modified Mann-Kendall test improves upon the original Mann-Kendall test by addressing some of its limitations, particularly regarding the sensitivity to the presence of autocorrelation in the data. Autocorrelation occurs when values in a time series are correlated with preceding or lagged values.

Assuming $(x_j - x_i) = \theta$, the value of sign(θ) is computed using equation 3.1.

$$\text{sign}(\theta) = \begin{cases} 1 & \text{if } \theta > 0 \\ 0 & \text{if } \theta = 0 \\ -1 & \text{if } \theta < 0 \end{cases} \qquad \text{Eq}^{\text{n}}\text{... 3.1}$$

The Sum of all sign value is calculated using equation 3.2.

$$S = \sum_{i=1}^{N-1} \sum_{j=i+1}^{N} \text{sign}(x_j - x_i), \qquad \text{Eq}^{\text{n}}\text{... 3.2}$$

where N is the number of data points.

S= Sum of all sign value.

This statistic represents the number of positive differences minus the number of negative differences for all the differences considered. For large samples (N > 10), the test is conducted using a normal distribution, with the mean and the variance.

The Variance of sum is calculated using equation 3.3.

$$\text{var}(S) = \left(\frac{N(N-1)(2N+5) - \sum_{k=1}^{n} t_k(t_k-1)(2t_k+5)}{18}\right), \qquad \text{Eq}^{\text{n}}\text{... 3.3}$$

Where;

N = Total Number

t_k = Number of terms whose difference (tied value) Zero.

Equation 3.4 is used for calculating Z Statistics.

$$Z = \begin{cases} \frac{S-1}{\sqrt{var(S)}} & \text{if } S > 0 \\ 0 & \text{if } S = 0 \\ \frac{s+1}{\sqrt{var(S)}} & \text{if } S < 0 \end{cases} \qquad \text{Eq}^{n}\ldots\ 3.4$$

Where, S= sum of data.

If the computed value of $|Z| > Z\alpha/2$, the null hypothesis (H_0) is rejected at α level of significance in a two-sided test. In this analysis, the null hypothesis was tested at 95% confidence level.

3.5.1.2 Sen's Slope test

Sen's approach which has been frequently used to assess the degree of trend in hydrometeorological time series (Lettenmaier et al.,1994; Yue and Hoshino, 2003; Partel and kanya, 2006), is based on assumption that the time series has a linear trend. Using this procedure, all data pairs slopes (T_i) are first determined using equation 3.5.

$$T_i = \frac{x_j - x_k}{j-k} \qquad \text{Eq}^{n}\ldots\ 3.5$$

$$\text{for I} = 1, 2, \ldots N$$

Where;

x_j and x_k are data values at time j and k (j>k), respectively.

The median of these N values of T_i is Sen's estimator of slope, which is calculated using equation 3.6.

$$\beta = \begin{cases} \frac{TN+1}{2} & N \text{ is odd} \\ \frac{1}{2}\left(\frac{TN}{2} + T\frac{N+2}{2}\right) & N \text{ is even} \end{cases} \qquad \text{Eq}^{n}\ldots\ 3.6$$

A positive value of β indicates an upward (increasing) trend and a negative value indicates a downward (decreasing) trend in the time series.

3.5.2 Morphometric Analysis

Morphometric analysis of a watershed involves evaluating its physical characteristics, including linear (stream order, length), areal (drainage density, form

factor), and relief aspects (basin relief, ruggedness). This analysis aids in flood risk assessment, soil erosion studies, and effective watershed management using GIS tools.

3.5.2.1 Stream order

Stream order is a classification of the hierarchy of streams within a watershed, indicating their relative size and connectivity. The formula for determining stream order follows Strahler's method:

1. First-order streams are the smallest, with no tributaries.
2. When two streams of the same order merge, the resulting stream is one order higher.
3. When streams of different orders merge, the resulting stream retains the higher order.

This helps in understanding the watershed's drainage network and hydrological behaviour.

3.5.2.2 Farm factor

The form factor is defined as the ratio between watershed area and the square of watershed length. It is a dimensionless number and will always be less than 1. This relationship is mathematically expressed by the equation 3.7. For a perfectly fan-shaped (circular) watershed, the numerical value of the form factor will be 0.786. Fern-shaped watersheds have values smaller than 0.786. A smaller numerical value of the form factor indicates a greater elongation of the watershed.

$$\text{Form factor} = \frac{A}{Lu^2} \qquad \text{Eq}^{n}\ldots\ 3.7$$

Where;

A = Area of drainage basin and

L = Overall Length of drainage basin.

3.5.2.3 Drainage density

Drainage density is a measure of how closely spaced stream channels are within a watershed. It is calculated by dividing the total length of all streams and rivers in the watershed by the total area of the watershed as mentioned in equation 3.8. High drainage density indicates numerous channels and rapid runoff, while low density

suggests fewer channels and slower runoff.

$$\text{Drainage density} = \frac{L_u}{A} \qquad \text{Eq}^n\ldots\ 3.8$$

Where;

Lu = Overall length of drainage basin and

A= Area of watershed.

3.5.2.4 Length of overland flow

The length of overland flow in a watershed is the average distance that water travels over the ground surface before reaching a stream channel as expressed in equation 3.9. It is typically measured from the point where rainfall hits the ground to the nearest stream. This parameter is important in hydrology as it affects the rate and volume of runoff, influencing flood potential and sediment transport.

$$\text{Length of overland flow} = \frac{1}{2D} \qquad \text{Eq}^n\ \ldots\ 3.9$$

Where;

D = Drainage density

3.5.2.5 Stream Frequency

Stream frequency, also known as channel frequency, is the number of stream channels per unit area within a watershed. This relationship is mathematically expressed in equation 3.10. It is calculated by dividing the total number of streams by the area of the watershed. High stream frequency indicates a dense network of streams, often associated with steep terrain and high runoff potential, while low stream frequency suggests a sparser network with less runoff.

$$\text{Stream frequency} = \frac{\text{No.of stream segment}}{\text{Area}} \qquad \text{Eq}^n\ldots 3.10$$

3.5.2.6 Elongation Ratio

Elongation ratio is defined as the ratio of diameter of a circle of the same area as the watershed to the maximum watershed length as shown in equation 3.11. The numerical value varies from 0 (in highly elongated shape) to 1 (in circular shape). These

values can be grouped in table 3.1 as,

Table 3.1: Numerical value of watershed

Elongation ratio	Shape of watershed
< 0.7	Elongated
0.8-0.7	Less elongated
0.9-0.8	Oval
>0.9	Circular

$$ER = \frac{2\sqrt{A/\pi}}{L} \qquad \text{Eq}^{n}\ldots\ 3.11$$

Where;

A= Area of the watershed, and

L= Maximum length.

A lower ER indicates a more elongated watershed, affecting hydrological response and flow concentration.

3.5.2.7 Circularity Ratio

Circulatory ratio is defined as the ratio of watershed area to the area of the circle having the same perimeter as the watershed perimeter as expressed in equation 3.12. The numeric value may vary in between 0 (in line) and1 (in a circle). In general, most of the watersheds have values from 0.2 to 0.8.

$$CR = \frac{4\pi A}{P^2} \qquad \text{Eq}^{n}\ldots\ 3.12$$

Where;

A = Area of the watershed, and

P = Perimeter.

A higher CR suggests a more circular shape, influencing runoff and erosion patterns.

Chapter 4

RESULTS AND DISCUSSION

This chapter offers a description of the methodology employed in the elevation, taking into account the many criteria that were calculated and the statistical relationship that was established between the criteria and the data. The final results that were attained after the whole analysis are included in this chapter.

4.1 Trend Analysis

The trend analysis of rainfall in Ambedkar Nagar, Uttar Pradesh, reveals significant fluctuations in precipitation patterns throughout the year. Z statistics values indicate deviations from the mean for different months, suggesting both above and below-average rainfall levels. Sen's values further support these findings, showing varying trends over different periods, including notable negative values during certain months, indicating decreasing trends in rainfall over time. A closer examination of Z statistics and Sen's values across different time frames highlights the complex dynamics of rainfall distribution in the region. Despite fluctuations, there is a clear indication of declining trends in certain periods, particularly during critical monsoon months.

4.1.1 Analysis of monthly Z statistics

The Z statistics monthly analysis is displayed in Fig. 4.1. The fact that no month trend value in the data is more than 1.96 indicates that there aren't any patterns that are obviously rising. A trend number greater than -1.96 indicates a steep decline in trend. Specifically, when the test statistic is more than -1.96 in June, July, August, September, November, and December, it exhibits markedly falling trends. Test findings for the following three months—January, May, and October—show no consistent trends across those intervals. In summary, there are a few months with all non-significant dropping tendencies in the data, but there are no major rising or increasing patterns.

4.1.2 Analysis of seasonal and annual Z statistics

According to the interpretation guidelines, a test statistic value above 1.96 indicates a significantly increasing trend, while a value below -1.96 suggests a significantly decreasing trend. Values between -1.96 and 1.96 indicate no significant trend. The pre-monsoon season has a test statistic of 3.09, indicating a significantly

increasing trend. The monsoon season has a test statistic of -0.43, suggesting no significant trend. The post-monsoon season has a test statistic of -1.36, and the winter season has a test statistic of -1.3866. Both values indicate significantly decreasing trends. The annual period has a test statistic of -3.34, indicating a significantly decreasing trend for the entire year as shown in Fig.4.2.

In summary, the data shows a significantly increasing trend during the pre-monsoon season, no significant trend during the monsoon season, and significantly decreasing trends during the post-monsoon, winter, and annual periods.

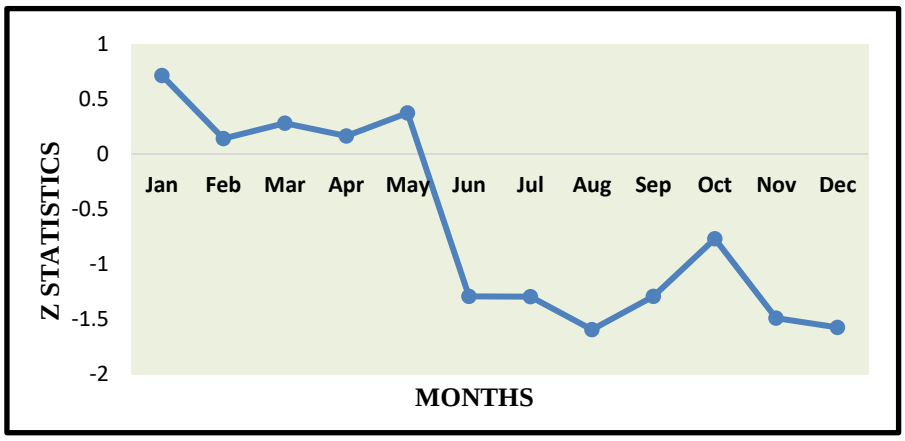


Fig 4.1 Monthly Z statistics

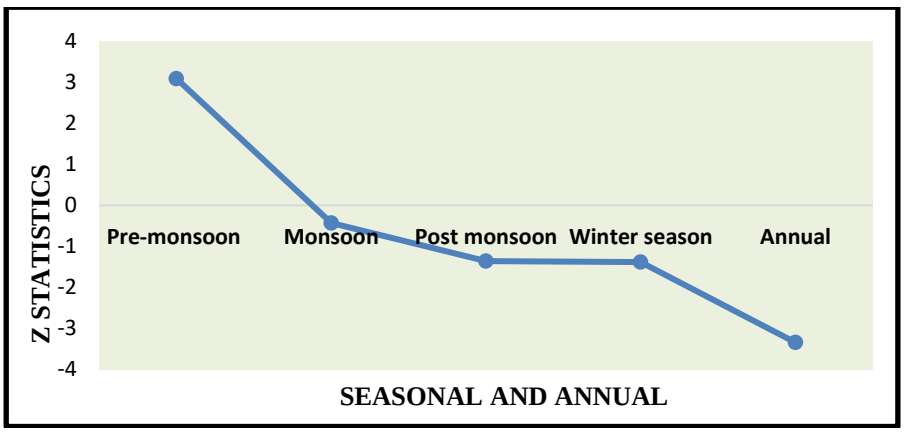


Fig 4.2 Seasonal and annual Z statistics

4.1.3 Decadal analysis of Z statistics

Analysing the Mann-Kendall test statistics across different months and decades reveals several trends. Decade 1 shows no significant trends, with all months' values within -1.96 to 1.96. Decade 2 displays pronounced decreasing trends, notably in June (-1.97), with most months having test statistics less than -1.96, indicating non-

significantly decreasing trends. In Decade 3, November stands out with a significant decreasing trend (-2.56), while other months show no significant trends. Decade 4 is balanced, with no months having test statistics beyond -1.96 or 1.96, suggesting no significant trends as shown in Fig.4.3.

Overall, significant decreasing trends are observed in specific months in Decades 2 and 3, particularly in September, February, June, July, August, and November. No significantly increasing trends appear in any decade. Decades 2 and 3 had pronounced decreasing trends, while Decade 4 shows stability with no significant trends.

For seasons and annual periods, Decade 01 shows no significant trends. Decade 02 has significant decreases in the Monsoon season (-2.86), Pre-monsoon season (-1.1), and Annual period (-2.5). Decade 03 indicates significant decreases in the Monsoon season (-1.43) and Annual period (-1.61). Decade 04 shows no significant trends. The middle decades (02 and 03) experienced notable declines, especially in the Monsoon season and Annual period, while Decades 01 and 04 exhibit stable patterns.

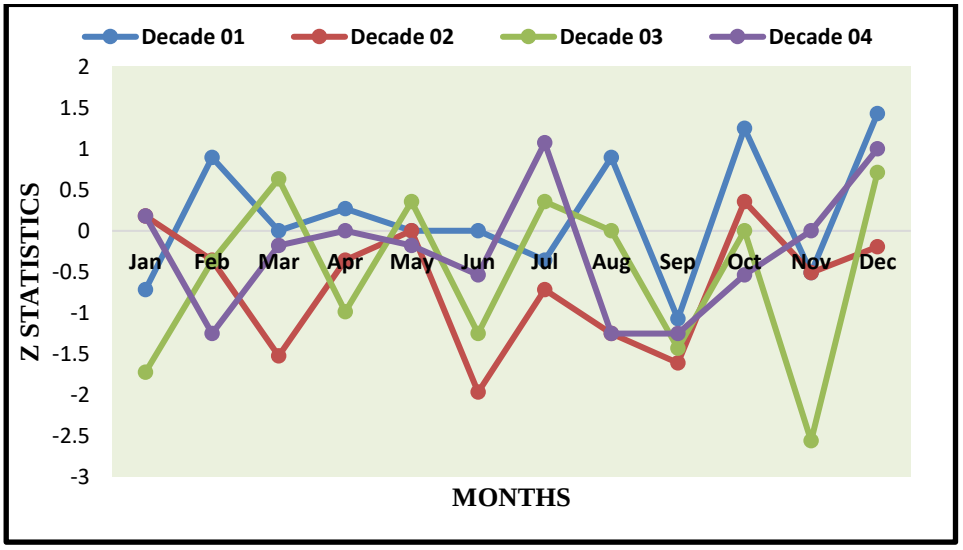


Fig 4.3 Decadal analysis of Z statistics

4.1.4 Seasonal and annual analysis in decade of Z statistics

Decade 01 shows no significant trends, with all test statistics between -1.96 and 1.96. In Decade 02, significant decreasing trends are observed in the Monsoon season (-2.86), Pre-monsoon season (-1.1), and Annual period (-2.5). Decade 03 indicates significant decreasing trends in the Monsoon season (-1.43) and Annual period (-1.61), though other seasons show no significant trends. Decade 04 again shows no significant trends, with all values within -1.96 to 1.96 as shown in Fig.4.4.

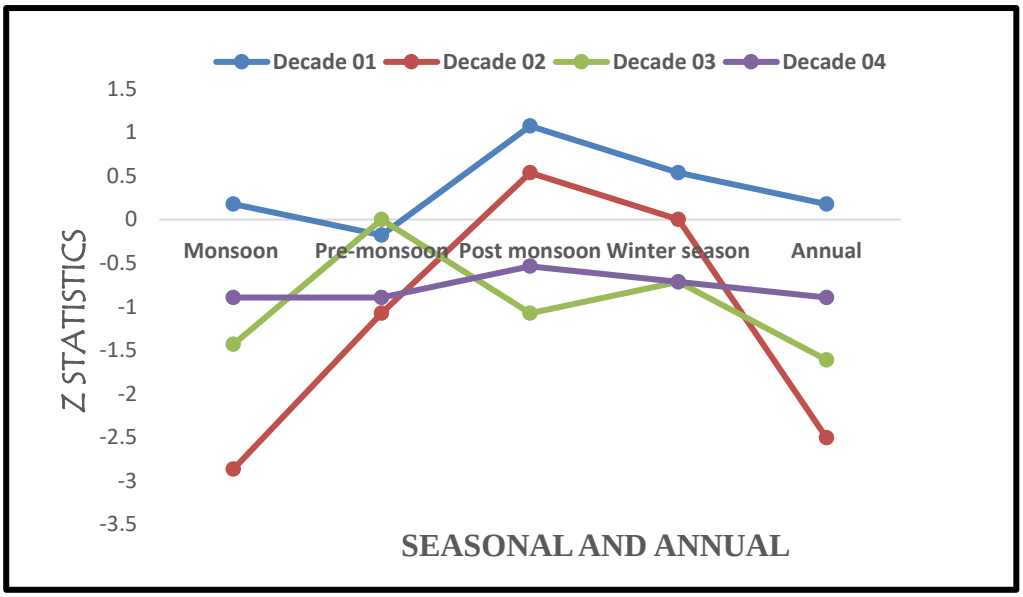


Fig 4.4 Seasonal and annual analysis in decade of Z statistics

4.1.5 Sen's value of monthly data

Analysing the trend magnitudes in Ambedkar Nagar, January, February, March, and April have stable trends (0.00). May shows a small decreasing trend (-0.11). Significant decreasing trends are observed in June (-1.57), July (-3.33), August (-2.25), and September (-3.38), the months with the largest decreases. October has a small decreasing trend (-0.19). November and December show stable trends (0.00) as shown in Table 4.1. The most significant decreasing trends occur during June to September. Other months exhibit smaller decreasing trends or remain stable. This analysis highlights seasonal variations and the relative strength of trends at the station over time.

Table 4.1: Sen's slope value 40 years

Sation	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Ambedkar Nagar	-0.05	0.00	0.00	0.00	-0.11	-1.57	-3.33	-2.25	-3.38	-0.19	0.00	-0.04
Increasing magnitude of trend				Zero					Decreasing magnitude of trend			

4.1.6 Sen's value of seasonal and annual analysis 40 years

Examining the trend magnitudes for Ambedkar Nagar reveals significant seasonal and annual decreases. The Monsoon season shows the largest decreasing trend (-11.18), indicating a substantial negative trend during these months. The Pre-monsoon season has a small decreasing trend (-0.17), while the post-monsoon (-0.34) and Winter

(-0.33) seasons show moderate decreasing trends. The Annual period has the most significant overall decrease (-12.50), the largest among all periods as shown in Table 4.2. All trend values are negative, indicating no increasing trends, with decreases ranging from small to significant. The data highlights substantial decreasing trends, particularly during the Monsoon season and annually. Other seasons also exhibit decreasing trends but are less pronounced compared to the Monsoon and Annual periods.

This analysis provides a comprehensive understanding of the trend strengths and directions, offering deeper insights into the climatic and environmental changes at the station over time.

Table 4.2: Seasonal and Annual Sen' Slope value of 40 years

Station	Monsoon	Pre-monsoon	Post-monsoon	Winter season	Annual
Ambedkar Nagar	-11.18	-0.17	-0.34	-0.33	-12.50
Increasing magnitude of trend		Zero		Decreasing magnitude of trend	

4.1.7 Sen's slope value of decadal analysis

Analyzing trend magnitudes across decades at Ambedkar Nagar reveals changing dynamics. In the first decade, moderate trends are seen, with August showing the largest increase (9.59) and September the most significant decrease (-11.58). The second decade shows severe decreasing trends, notably in September (-29.56), June (-22.01), and August (-17.13), indicating worsening conditions. The third decade continues with significant decreases in September (-22.13) and June (-17.04), while July (5.27) shows a substantial increase. The fourth decade has a mix of trends, with July (14.74) increasing and September (-12.21), August (-10.46), and May (-2.10) decreasing as shown in Table 4.3.

Overall, the second decade stands out for its severe decreasing trends, especially in September. Other decades show a mix of trends, but the general pattern indicates a shift towards less favourable conditions. Understanding these changes' underlying factors and implications for the local environment, economy, and community is crucial, necessitating further research and analysis.

Table 4.3: Sen's slope value of decadal analysis

Station	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Decade-1	-1.25	0.65	-0.02	0.20	-0.04	4.21	-5.56	9.59	-11.58	5.63	0.00	0.95
Decade-2	0.20	-0.80	-0.48	-0.08	0.00	-22.01	-11.48	-17.13	-29.56	1.11	0.00	0.00
Decade-3	-0.17	0.00	0.04	-0.21	0.91	-17.04	5.27	0.76	-22.13	-0.20	-0.16	0.00
Decade-4	0.22	-1.19	-0.10	-0.16	-2.10	-1.37	14.74	-10.46	-12.21	-1.05	0.00	0.00
Increasing magnitude of trend				Zero				Decreasing magnitude of trend				

4.1.8 Sen's value of seasonal and annual decadal analysis

Examining trend magnitudes across seasons for each decade at Ambedkar Nagar reveals significant changes. In the first decade, trends were moderate with a positive Monsoon trend (8.65) and an annual trend (17.08) indicating improvement. The second decade saw sharp deterioration, with an extreme Monsoon decrease (-80.07) and a strongly negative Annual trend (-87.89). The third decade continued this decline, particularly in the Monsoon season (-38.28), though less severe than the second decade as shown in Table 4.4.

The fourth decade showed slight improvement but remained challenging, with a Monsoon decrease (-20.95) and an overall negative Annual trend (-36.05).

Across four decades, the station experienced worsening conditions, especially during the Monsoon season. The second decade was the most severe, with significant impacts on the annual trend. Despite some moderation in later decades, significant challenges persist, highlighting the need for further investigation and intervention to address these changes.

Table 4.4: Table of seasonal and annual decadal analysis

Station	Monsoon	Pre-monsoon	Post monsoon	Post monsoon	Annual
Decade01	8.65	-0.32	5.19	1.94	17.08
Decade02	-80.07	-1.15	2.29	-0.16	-87.89
Decade03	-38.28	-1.13	-2.39	-1.89	-43.94

Decade04	-20.95	-3.28	-1.05	-1.05	-36.05
Increasing magnitude of trend		Zero		Decreasing magnitude of trend	

4.2 Morphometric Analysis

The morphometric analysis of Ambedkar Nagar District, encompassing a total geographical area of 2529.68 square kilometers, revealed compelling insights into its hydrological features. With a total stream length of 975,818 meters, the district showcases a complex network of watercourses. By dissecting the main stream flow path, significant patterns in the drainage system emerge, shedding light on the district's hydrological dynamics. These findings not only provide valuable information for understanding the region's water resource management but also offer crucial insights for sustainable development initiatives and environmental conservation efforts in Ambedkar Nagar District.

4.2.1 Stream order

The Maharuwa micro-watershed has 14,465 first-order, 6,983 second-order, 2,752 third-order, 4,420 fourth-order, and 320 fifth-order streams as shown in table 4.5. Notably, fourth-order streams are unusually numerous, suggesting unique local conditions. This data is crucial for hydrological studies, watershed management, and environmental planning in the region.

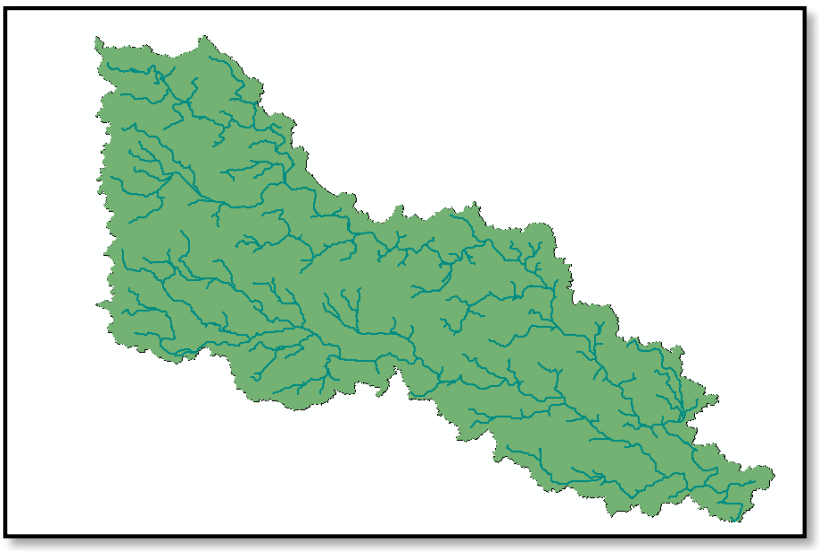


Fig 4.5: Streams of Maharuwa micro-watershed

Table 4.5: Stream orders and no. of stream order.

Serial no.	Stream order	No. of streams
1.	1	14465
2.	2	6983
3.	3	2752
4.	4	4420
5.	5	320

4.2.2 Form factor

Form factor is a ratio of watershed area to the square of the length of the watershed.

$$\text{Form factor} = \frac{A}{Lu^2}$$

$$= \frac{2529.68}{(975.818)^2}$$

$$= 0.0027$$

Where, A = Area of drainage basin; L = overall Length of drainage basin

In the Maharuwa micro-watershed the value of form factor is 0.0027.

4.2.3 Drainage density

Drainage density is a measurement of the sum of the channel lengths per unit area.

$$\text{Drainage density} = \frac{L_u}{A}$$

$$= \frac{975.818}{2529.68}$$

$$= 0.38574 \text{ km/square km}$$

Where, Lu = Overall length of drainage basin; A= Area

The drainage density in the Maharuwa micro-watershed is 0.38574 km/square

kilometre.

4.2.4 Length of overland flow

The length of overland flow is determined by the maximum distance rainwater flows on a drainage-basin surface before reaching a channel.

$$\text{Length of overland flow} = \frac{1}{2D}$$

$$= \frac{1}{2 \times 0.385}$$

=1.2962 km

Where, D = Drainage density

1.2962 kilometres is the length of the overland flow in the Maharuwa micro-watershed.

4.2.5 Stream Frequency

The stream frequency is the ratio between the total number of stream segments of all orders in a watershed and the basin/watershed area.

$$\text{Stream frequency} = \frac{\text{No. of stream segment}}{\text{Area}}$$

$$= \frac{28940}{2529.68}$$

=11.44 per square kilometer

Stream frequency is 11.44 per square km at the Maharuwa micro-watershed.

4.2.6 Elongation Ratio

Elongation ratio is defined as the ratio of diameter of a circle of the same area as the watershed to the maximum watershed length.

$$ER = \frac{2\sqrt{A/\pi}}{L}$$

$$= \frac{2\sqrt{(2529.68/3.14)}}{975.818}$$

$$= 0.058174$$

Where, A= Area of the watershed, and L= Maximum length.

In the Maharuwa micro-watershed the value of elongation ratio is 0.058174.

4.2.7 Circularity Ratio

Circulatory ratio is defined as the ratio of watershed area to the area of the circle having the same perimeter as the watershed perimeter.

$$CR = \frac{4\pi A}{P^2}$$

$$= \frac{4 \times 3.14 \times 2529.68}{178.06^2}$$

$$=1.002$$

Where, A = Area of the watershed, and P = Perimeter.

In the Maharuwa micro-watershed the value of circularity ratio is 1.002.

Table 4.6: Morphometric parameters and their values.

Sr. No.	**Parameters**	**Values**
1.	Form factor	0.0027
2.	Drainage density	0.38574 km/square km
3.	Length of overland flow	1.2962 km
4.	Stream frequency	11.44 per square kilometer
5.	Elongation Ratio	0.058174
6.	Circularity Ratio	1.002

Chapter 5

SUMMARY AND CONCLUSIONS

This thesis addresses two key objectives. The first is to analyze the trend of rainfall in Ambedkar Nagar over a 40-year period, from 1980 to 2019. By utilizing historical rainfall data and applying statistical methods, the study identifies significant patterns, trends, and anomalies in precipitation levels. The analysis examines changes in rainfall distribution, seasonal variations, and potential impacts of climate change, providing essential insights for effective water resource management and agricultural planning.

The second objective is to estimate the morphometric parameters of the Maharuwa micro-catchment located in Ambedkar Nagar. Morphometric analysis involves the quantitative evaluation of the catchment's physical characteristics, such as its drainage network, shape, and relief aspects. This assessment helps in understanding the hydrological behavior and potential for soil erosion within the catchment. The study employs GIS techniques to accurately measure parameters like stream order, drainage density, and basin shape.

Together, these analyses offer a comprehensive understanding of the climatic and hydrological dynamics of Ambedkar Nagar. The insights gained are crucial for local stakeholders and policymakers to develop adaptive strategies for sustainable water management, agricultural productivity, and environmental conservation in the region.

- From the analysis of rainfall data of 40 years, significantly increasing trend is found during pre-monsoon season.
- A notable downward trend is observed during the monsoon and post-monsoon seasons based on a 40-year review of rainfall data.
- In the Maharuwa micro-catchment 0.38 km length was calculated per square km of basin area.
- Form factor was estimated as 0.0027 for Maharuwa micro-catchment.

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ABSTRACT

The analysis of rainfall trends and their magnitudes in Ambedkar Nagar District, Uttar Pradesh, is critical for understanding the region's climatic patterns and planning for agricultural and water resource management. This study utilizes historical rainfall data spanning from (1980-2019) and four decades to identify and quantify trends using statistical methods such as the Mann-Kendall test, Sen's slope estimator. In pre-monsoon have positive trend and others all season and annual have decreasing trend. The results reveal significant patterns in annual and seasonal rainfall, providing insights into potential changes in precipitation that could impact local agriculture, water supply, and flood management strategies. By quantifying the rate of change in rainfall, the study aims to assist policymakers and stakeholders in developing adaptive measures to mitigate the effects of climate variability.

The study also focuses on the estimation of morphometric parameters of the Maharuwa micro-watershed, an essential aspect of understanding the hydrological behaviour of the region. Using Digital Elevation Models (DEM) and GIS tools, various morphometric parameters such as catchment area (2529.68), perimeter (178.06), fifth order stream, farm factor (0.0027), drainage density (0.385), elongation ratio (0.0581), circulatory ratio (1.002) and length of overland flow (1.296) are calculated. These parameters provide a comprehensive understanding of the catchment's physical characteristics, which are crucial for effective watershed management and planning. The morphometric analysis helps in assessing the catchment's potential for runoff, soil erosion, and flood susceptibility, thereby informing sustainable water resource management practices in the Maharuwa micro-watershed area.

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